

How to install ansible in AWS EC2:

Requirements: VPC with **2 subnets** (public and private in different Availability Zone) and **2 EC2s** that runs **ubuntu** on it in the public subnet.

Note:

 **Master:** is referred to the machine that will deploy the ansible playbook (Control Node)

 **Slave:** is the **target machine** that Ansible configures (Managed/Client Node)

Step 1: update the EC2 by running the command “**Sudo apt update**”

Step 2: run these commands in the master machine to **install ansible**:

-  `sudo apt install software-properties-common`
-  `sudo add-apt-repository --yes --update ppa:ansible/ansible`
-  `sudo apt install ansible`

step 3: to check that you install ansible right issue the command “**ansible --version**”

```
ubuntu@ip-192-168-1-23: ~  
Setting up python3-xmltodict (0.13.0-1) ...  
Setting up ansible (11.5.0-1ppa~noble) ...  
Setting up python3-nacl (1.5.0-4build1) ...  
Setting up python3-requests-ntlm (1.1.0-3) ...  
Setting up python3-winrm (0.4.3-2) ...  
Setting up python3-paramiko (2.12.0-2ubuntu4.1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-192-168-1-23:~$ ansible --version  
ansible [core 2.18.5]  
  config file = /etc/ansible/ansible.cfg  
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']  
  ansible python module location = /usr/lib/python3/dist-packages/ansible  
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections  
  executable location = /usr/bin/ansible  
  python version = 3.12.3 (main, Feb  4 2025, 14:48:35) [GCC 13.3.0] (/usr/bin/python3)  
  jinja version = 3.1.2  
  libyaml = True  
ubuntu@ip-192-168-1-23:~$ |
```

Now you can check the ansible directory if you go to “**cd /etc/ansible**”

```
ubuntu@ip-192-168-1-23: /etc  x  ubuntu@ip-192-168-1-64: ~  x  +  v
ubuntu@ip-192-168-1-23:~$ cd /etc/ansible/
ubuntu@ip-192-168-1-23:/etc/ansible$ ls
ansible.cfg  hosts  roles
ubuntu@ip-192-168-1-23:/etc/ansible$ |
```

As you can see there are **3 files in the ansible directory**, but you will use two:

- 📄 **ansible.cfg** is the file that has the configuration for the ansible
- 📄 **hosts** where you list your Ip address for the servers or the machine that you want to remotely configure.

```
ubuntu@ip-192-168-1-23: /etc  x  ubuntu@ip-192-168-1-64: ~  x  +  v
ubuntu@ip-192-168-1-23:~$ cd /etc/ansible/
ubuntu@ip-192-168-1-23:/etc/ansible$ ls
ansible.cfg  hosts  roles
ubuntu@ip-192-168-1-23:/etc/ansible$ ssh ubuntu@192.168.1.64
The authenticity of host '192.168.1.64 (192.168.1.64)' can't be established.
ED25519 key fingerprint is SHA256:J+soTBIEG9lpkYBFL3afDfCTrDeq9D3wWcK/Cm9UHvE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.64' (ED25519) to the list of known hosts.
ubuntu@192.168.1.64: Permission denied (publickey).
ubuntu@ip-192-168-1-23:/etc/ansible$ |
```

step 4: To access the slave machine via SSH, the slave must have the public key of the master machine.

```
ubuntu@ip-192-168-1-23: /etc/ X  ubuntu@ip-192-168-1-64: ~/s X + v
ubuntu@ip-192-168-1-64:~$ cd .ssh
ubuntu@ip-192-168-1-64:~/.ssh$ ls
authorized_keys
ubuntu@ip-192-168-1-64:~/.ssh$ |
```

Step 5: you have to generate a rsa key from the master machine

First: go to the **master machine** in the home directory the go to .ssh file by issue the command “cd .ssh”

Note: if you don’t know were you are to go back to the home directory issue the command “cd ~” and you will be in the home directory.

```
ubuntu@ip-192-168-1-23: ~/s X  ubuntu@ip-192-168-1-64: ~/ss X + v
ubuntu@ip-192-168-1-23:~$ cd .ssh
ubuntu@ip-192-168-1-23:~/.ssh$
ubuntu@ip-192-168-1-23:~/.ssh$ ls
authorized_keys  known_hosts
ubuntu@ip-192-168-1-23:~/.ssh$
```

If you enter “**ls**” command, you will see that the .ssh folder has one file the authorized_keys.

```
ubuntu@ip-192-168-1-23: /etc/ X ubuntu@ip-192-168-1-64: ~/s X + v
ubuntu@ip-192-168-1-64: ~/.ssh$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:81XAiSwoPG7wmY9Th1MGYKHD9wDHBXMAiJopuoFd5b4 ubuntu@ip-192-168-1-64
The key's randomart image is:
+--[ED25519 256]--+
|..o+O*+o . o..|
|o..==oo + o o.|
|.o++oB + . .|
|= oBo= . .|
|+. o =.oS .|
|+ . o o o .|
|o . . .|
|. E|
+-----[SHA256]-----+
ubuntu@ip-192-168-1-64: ~/.ssh$ ls
authorized_keys id_ed25519 id_ed25519.pub
ubuntu@ip-192-168-1-64: ~/.ssh$
```

Second: run the command in the Master machine “ssh-keygen” then **press enter 3 time** to generate the key that you will use for ssh into the slave machine.

To check if you generate the key correctly enter “ls” you should have 3 files two of them like this “id_.....” and “id_-----.pub” you need the .pub key for the ssh to work.

```
ubuntu@ip-192-168-1-23: /etc/ X ubuntu@ip-192-168-1-64: ~/s X + v
ubuntu@ip-192-168-1-64: ~/.ssh$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:81XAiSwoPG7wmY9Th1MGYKHD9wDHBXMAiJopuoFd5b4 ubuntu@ip-192-168-1-64
The key's randomart image is:
+--[ED25519 256]--+
|..o+O*+o . o..|
|o..==oo + o o.|
|.o++oB + . .|
|= oBo= . .|
|+. o =.oS .|
|+ . o o o .|
|o . . .|
|. E|
+-----[SHA256]-----+
ubuntu@ip-192-168-1-64: ~/.ssh$ ls
authorized_keys id_ed25519 id_ed25519.pub
ubuntu@ip-192-168-1-64: ~/.ssh$ cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPQpSr+G5hP4g6sg/GAcGtwPckNJF6ge8ikt7GTtMSED ubuntu@ip-192-168-1-64
ubuntu@ip-192-168-1-64: ~/.ssh$
```

Third: copy the .pub by using the command “cat <id.pem>” and copy it.

```

GNU nano 7.2 authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADA0ABAAQACbLqKECxi5wia/9bohZ2sGwYo0IicGz+D4fE7wqLtox4zRLL+mGVdj5CEIE0qT4ApIPytCtn8ze8C80N/
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPQpSr+G5hP4g6sg/GAcGtwPckNJF6ge8ikt7GTtMSED ubuntu@ip-192-168-1-64
[ Read 2 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^J Paste      ^_ Justify    ^_ Go To Line M-E Redo      M-6 Copy

```

Fourth: go to your **slave machine** and enter “**cd .ssh**” then “**ls**” to list what in the folder you will find the “**authorized_keys**” file open it using this command “**sudo nano authorized_keys**” then in the bottom **paste the key** you copy from the Master machine.

Note: to save and Exit click on “CTRL X then Y then press Enter”.

```

ubuntu@ip-192-168-1-23: ~/.ssh$ ls
authorized_keys  id_ed25519  id_ed25519.pub  known_hosts
ubuntu@ip-192-168-1-23: ~/.ssh$ sudo nano authorized_keys
ubuntu@ip-192-168-1-23: ~/.ssh$ ssh ubuntu@3.88.62.77
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1024-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri May  2 21:59:35 UTC 2025

System load:  0.0          Processes:      113
Usage of /:   35.7% of 6.71GB Users logged in: 1
Memory usage: 30%         IPv4 address for enX0: 192.168.1.64
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

84 updates can be applied immediately.
38 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri May  2 21:51:51 2025 from 3.92.45.227
ubuntu@ip-192-168-1-64:~$ |

```

Now you can check that you done everything by ssh to the slave machine

Enter the command “**ssh ubuntu@<the public ip for the slave machine>**”

As you can see, the ssh works.

Ansible Configuration:

```
ubuntu@ip-192-168-1-23: /etc  ×  ubuntu@ip-192-168-1-64: /etc/  ×  +  v
ubuntu@ip-192-168-1-23:~$ cd /etc/ansible/
ubuntu@ip-192-168-1-23:/etc/ansible$ ls
ansible.cfg  hosts  roles
ubuntu@ip-192-168-1-23:/etc/ansible$ |
```

Now we have to configure ansible go to the **master machine** in the ansible folder that located in the “**etc**” you can go there by issue the command “**cd /etc/ansible/**”

```
ubuntu@ip-192-168-1-23: /etc  ×  ubuntu@ip-192-168-1-64: /etc/  ×  +  v
GNU nano 7.2                                ansible.cfg
# Since Ansible 2.12 (core):
# To generate an example config file (a "disabled" one with all default settings, commented out):
#     $ ansible-config init --disabled > ansible.cfg
#
# Also you can now have a more complete file by including existing plugins:
# ansible-config init --disabled -t all > ansible.cfg
#
# For previous versions of Ansible you can check for examples in the 'stable' branches of each version
# Note that this file was always incomplete and lagging changes to configuration settings
# for example, for 2.9: https://github.com/ansible/ansible/blob/stable-2.9/examples/ansible.cfg

[ Read 11 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  ^U Undo      ^A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line ^E Redo      ^G Copy
```

Step six: open the ansible.cfg by entering the command “sudo nano ansible.cfg” it will open like this

Copy the configuration and paste it in the file:

[defaults]

inventory = /etc/ansible/hosts

remote_user = **ubuntu** ***change it to the username of your machine**

ask_pass = false

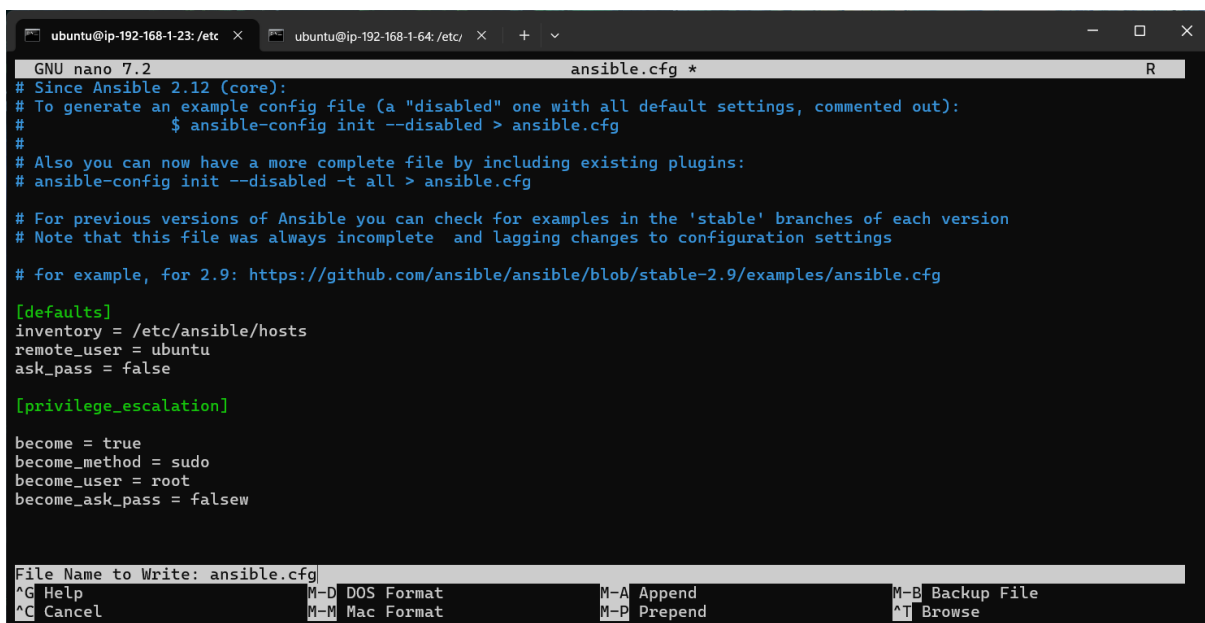
[privilege_escalation]

become = true

become_method = sudo

become_user = root

become_ask_pass = false



```
GNU nano 7.2 ansible.cfg *
# Since Ansible 2.12 (core):
# To generate an example config file (a "disabled" one with all default settings, commented out):
#     $ ansible-config init --disabled > ansible.cfg
#
# Also you can now have a more complete file by including existing plugins:
# ansible-config init --disabled -t all > ansible.cfg
#
# For previous versions of Ansible you can check for examples in the 'stable' branches of each version
# Note that this file was always incomplete and lagging changes to configuration settings
# for example, for 2.9: https://github.com/ansible/ansible/blob/stable-2.9/examples/ansible.cfg

[defaults]
inventory = /etc/ansible/hosts
remote_user = ubuntu
ask_pass = false

[privilege_escalation]

become = true
become_method = sudo
become_user = root
become_ask_pass = false

File Name to Write: ansible.cfg
^G Help          M-D DOS Format   M-A Append       M-B Backup File
^C Cancel        M-M Mac Format   M-P Prepend      ^T Browse
```

Exit and save the changes.

```

ubuntu@ip-192-168-1-23: /etc  x  ubuntu@ip-192-168-1-64: /etc/  x  +  v
ubuntu@ip-192-168-1-23:~$ cd /etc/ansible/
ubuntu@ip-192-168-1-23:/etc/ansible$ ls
ansible.cfg  hosts  roles
ubuntu@ip-192-168-1-23:/etc/ansible$ sudo nano ansible.cfg
ubuntu@ip-192-168-1-23:/etc/ansible$ cat ansible.cfg
# Since Ansible 2.12 (core):
# To generate an example config file (a "disabled" one with all default settings, commented out):
# $ ansible-config init --disabled > ansible.cfg
#
# Also you can now have a more complete file by including existing plugins:
# ansible-config init --disabled -t all > ansible.cfg

# For previous versions of Ansible you can check for examples in the 'stable' branches of each version
# Note that this file was always incomplete and lagging changes to configuration settings

# for example, for 2.9: https://github.com/ansible/ansible/blob/stable-2.9/examples/ansible.cfg

[defaults]
inventory = /etc/ansible/hosts
remote_user = ubuntu
ask_pass = false

[privilege_escalation]

become = true
become_method = sudo
become_user = root
become_ask_pass = false
ubuntu@ip-192-168-1-23:/etc/ansible$ |

```

You can check if you save the changes by entering “**cat ansible.cfg**”

```

ubuntu@ip-192-168-1-23: /etc  x  ubuntu@ip-192-168-1-64: /etc/  x  +  v
GNU nano 7.2                                hosts
# This is the default ansible 'hosts' file.
#
# It should live in /etc/ansible/hosts
#
# - Comments begin with the '#' character
# - Blank lines are ignored
# - Groups of hosts are delimited by [header] elements
# - You can enter hostnames or ip addresses
# - A hostname/ip can be a member of multiple groups
#
# Ex 1: Ungrouped hosts, specify before any group headers:
## green.example.com
## blue.example.com
## 192.168.100.1
## 192.168.100.10
#
# Ex 2: A collection of hosts belonging to the 'webservers' group:
## [webservers]
## alpha.example.org
## beta.example.org
## 192.168.1.100
## 192.168.1.110
#
# If you have multiple hosts following a pattern, you can specify
[ Read 54 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line  M-E Redo      M-6 Copy

```

Step eight: open the hosts file by suing “**sudo nano hosts**” then copy the private ip for the slave machine.

Instances (1/2) [Info](#)

Find Instance by attribute or tag (case-sensitive) Running Last updated less than a minute ago Connect Instance state Actions Launch instances

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐	Master	i-0a15fba57ffc7203	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	-	3.92.45.227	-	-
☒	Slave	i-0d912bb0dce718781	Running	t2.micro	2/2 checks passed	View alarms	us-east-1a	-	3.88.62.77	-	-

i-0d912bb0dce718781 (Slave)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ Instance summary [Info](#)

Instance ID
[i-0d912bb0dce718781](#)

IPv6 address
-

Hostname type
IP name: ip-192-168-1-64.ec2.internal

Answer private resource DNS name
-

Auto-assigned IP address
[3.88.62.77](#) [Public IP]

Public IPv4 address
[3.88.62.77](#) [open address](#)

Instance state
[Running](#)

Private IP DNS name (IPv4 only)
[ip-192-168-1-64.ec2.internal](#)

Instance type
t2.micro

VPC ID
[vpc-0475de0c00078be19 \(Hussain-VPC\)](#)

Private IPv4 addresses
[192.168.1.64](#)

Public IPv4 DNS
-

Elastic IP addresses
-

AWS Compute Optimizer finding
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```

ubuntu@ip-192-168-1-23: /etc  x  ubuntu@ip-192-168-1-64: /etc/  x  +  v
GNU nano 7.2                                hosts *
# You can also use ranges for multiple hosts:

## db-[99:101]-node.example.com

# Ex 3: A collection of database servers in the 'dbservers' group:

## [dbservers]
##
## db01.intranet.mydomain.net
## db02.intranet.mydomain.net
## 10.25.1.56
## 10.25.1.57

# Ex4: Multiple hosts arranged into groups such as 'Debian' and 'openSUSE':

## [Debian]
## alpha.example.org
## beta.example.org

## [openSUSE]
## green.example.com
## blue.example.com

[ubuntu]
192.168.1.64
Save modified buffer?
Y Yes
N No      ^C Cancel
  
```

in the bottom of the file enter a name for the ip that you want to manage like [ubuntu] . Then under it enter the ip of the slave machine you copy it then exit and save the changes.

```
ubuntu@ip-192-168-1-23:/etc$ ansible all -m ping
[WARNING]: Platform linux on host 192.168.1.64 is using the discovered Python interpreter at /usr/bin/python3.12, but
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.18/reference_appendices/interpreter_discovery.html for more information.
192.168.1.64 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@ip-192-168-1-23:/etc/ansible$ |
```

Step nine: to check if you done everything correctly you will ping the slave machine
By entering the command “ansible all -m ping”.

✓ If you have done everything right, it will give you SUCCESS massage.

```
GNU nano 7.2 hosts
## db-[99:101]-node.example.com
# Ex 3: A collection of database servers in the 'dbservers' group:
## [dbservers]
##
## db01.intranet.mydomain.net
## db02.intranet.mydomain.net
## 10.25.1.56
## 10.25.1.57

# Ex4: Multiple hosts arranged into groups such as 'Debian' and 'openSUSE':
## [Debian]
## alpha.example.org
## beta.example.org

## [openSUSE]
## green.example.com
## blue.example.com

[ubuntu]
slave ansible_ssh_host=192.168.2.31 ansible_python_interpreter=/usr/bin/python3.12
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^/_ Go To Line M-E Redo     M-6 Copy
```

To remove the warning and give it a name for the client machine in the hosts file.